Unit 4, Lesson 10: Applying Multiple Laws of Exponents – Part 1

Benchmark

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Learning Target

- ☐ **I can write an equivalent algebraic expression by applying multiple laws of exponents.**

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4.10.1 Warm-Up: Crack the Safe

1. To crack the safe, you must fill in the middle columns with numbers from the bank. Each number can only be used once. Each row must add to equal 18. You have 5 minutes before it is locked forever.

Number Bank									
8	4	5	2	9	6	7	13	1	3

Code			Row Total
3			18
6			18
2			18
9			18
12			18

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4.10.2 Guided Practice: Applying Multiple Laws of Exponents to Algebraic Expressions

The laws of exponents are included in the bank below for you to reference as you complete the following problems.

Laws of Exponents

product of powers	quotient of powers	power of a power
power of a product	power of a quotient	negative exponent
identity exponent	zero exponent	

1. **Miguel and Gia generated the same expression equivalent to $(r^3 \cdot r^2)^2$. They both applied laws of exponents, but in different ways. Their work is shown.**

Miguel	Gia
$(r^3 \cdot r^2)^2$ $(r^5)^2$ r^{10}	$(r^3 \cdot r^2)^2$ $(r^6 \cdot r^4)$ r^{10}

   a. **For each student, identify which laws of exponents they applied and in what order.**

Order of Laws Applied	Miguel	Gia
First		
Second		

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b. Discuss with your partner which strategy you prefer. Summarize your discussion.

2. Jack and Angel rewrote the expression $\left(\frac{x^3y^{-4}}{xy^2}\right)^2$ in different ways and arrived at different expressions. Their work is shown.

Jack	Angel
$\left(\frac{x^3y^{-4}}{xy^2}\right)^2$	$\left(\frac{x^3y^{-4}}{xy^2}\right)^2$
$(x^2y^{-2})^2$	$\frac{x^6y^{-8}}{x^2y^4}$
x^4y^{-4}	x^4y^{-12}
$\frac{x^4}{y^4}$	$\frac{x^4}{y^{12}}$

a. Even though one of the student's resulting expressions is not equivalent to $\left(\frac{x^3y^{-4}}{xy^2}\right)^2$, identify which laws of exponents each student applied and in what order.

Order of Laws Applied	Jack	Angel
First		
Second		
Third		

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b. Which student generated an expression that is not equivalent to $\left(\frac{x^{3}y^{-4}}{xy^{2}}\right)^{2}$?

c. Which statement best describes the incorrect student's strategy?

- o The incorrect student used a correct strategy, but applied one of the laws incorrectly.
- o The incorrect student tried to use an exponent law that didn't apply to this expression.

d. Based on your response to Part C, show how to correct the student's error.

$$\left(\frac{x^{3}y^{-4}}{xy^{2}}\right)^{2}$$

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4.10.3 Try It: Applying Multiple Laws of Exponents to Algebraic Expressions

1. For each expression, write an equivalent expression using positive exponents and the fewest factors possible. Show your work in the space provided. Then, for each expression, check off the exponent laws applied from the list provided.

   a. $\left(\frac{9c^{-4}}{c^{-4}}\right)^2$

Exponent Laws Applied	
☐	identity exponent
☐	negative exponent
☐	power of a power
☐	power of a product
☐	power of a quotient
☐	product of powers
☐	quotient of powers
☐	zero exponent

   b. $(d^0 \cdot d^5)^{-2}$

Exponent Laws Applied	
☐	identity exponent
☐	negative exponent
☐	power of a power
☐	power of a product
☐	power of a quotient
☐	product of powers
☐	quotient of powers
☐	zero exponent

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4.10.4 Activity: Exponent Laws Relay

Work with your partner to complete the relay.

- **Start by generating an equivalent expression for question 1 using the fewest factors possible.**
- **Insert the answer from question 1 into the box in question 2. Repeat until you have completed all of the questions in this manner.**

1. $\frac{5x^{-6}x^{-2}}{25x^{-4}}$ = ______________

2. $10x^{3} \cdot \boxed{}$ = ______________

3. $\frac{(2x^{-4}x^{3})^{2}}{\boxed{}}$ = ______________

4. $\left(\frac{x}{2x^{4}}\right)^{2} \cdot \boxed{}$ = ______________

5. $x^{3} \cdot \boxed{}$ = ______________

To successfully complete the relay, your final answer should be 1. Go back to check your answers if needed.

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4.10.5 Your Turn!

1. For each expression, write an equivalent expression using positive exponents and the fewest factors possible. Show your work in the space provided. Then, for each expression check off the exponent laws applied from the list provided.

   a. $\frac{8x^{\text{[illegible]}}}{2x^{\text{[illegible]}}} \cdot x^{4}$

Exponent Laws Applied	
☐	identity exponent
☐	negative exponent
☐	power of a power
☐	power of a product
☐	power of a quotient
☐	product of powers
☐	quotient of powers
☐	zero exponent

   b. $\left(\frac{p^{\text{[illegible]}} \cdot q^{-\text{[illegible]}}}{(p^{-\text{[illegible]}} \cdot q)^{\text{[illegible]}}}\right)$

Exponent Laws Applied	
☐	identity exponent
☐	negative exponent
☐	power of a power
☐	power of a product
☐	power of a quotient
☐	product of powers
☐	quotient of powers
☐	zero exponent

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4.10.6 Wrap-Up: Ticket Out the Door

1. Write an equivalent expression using positive exponents and the fewest factors possible.

$$\left(\frac{3m^3n^2}{6mn^4}\right)^2$$

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